

THE IMPACT OF ESG SCORES ON S&P 500 STOCK PERFORMANCE

*A Data-Driven Analysis of Environmental, Social & Governance
Factors in Short-Term Returns*

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Business Question

Do ESG (Environmental, Social, Governance) scores have a meaningful impact on short-term stock performance for S&P 500 companies?

Specifically, we investigate:

- Which variables, ESG pillars or traditional financial metrics, best predict 1-year returns?
- Are certain sectors (e.g. Real Estate) uniquely affected by ESG factors?
- Does ESG interact with financial risk (Beta) to amplify stock performance?

Data Overview

Dataset

- **416 S&P 500 companies**
- 22 original variables covering ESG scores, financial metrics, and sector classifications
- ESG data sourced from Sustainalytics risk ratings
- Financial data from Yahoo Finance and market indices
- 385 companies after cleaning missing values

Target Variable

1 Year Return

1-year stock return measured from September 1, 2023 to September 1, 2024

**S&P 500 benchmark return during this period:
24%**

For classification models, stocks returning above 24% were labeled as "beat market"

Data Cleaning & Preprocessing

Dropped for Data Leakage

2023 and 2024 stock prices removed — returns are calculated directly from prices, so including them would leak the answer into the model.

Removed Redundancy

Combined ESG dropped because it is simply the sum of E + S + G. Keeping the three pillars separately allows us to see which dimensions matter most.

Handled High-Cardinality Categoricals

Sub Industries (119 categories) and Industry (107 categories) were dropped. Sector (11 categories) was dummy-encoded, producing 10 manageable dummy variables.

Missing Values

31 rows with missing values, or were no longer on the S&P in 2024 were dropped, leaving 385 clean observations for modeling.

Feature Selection: Forward, Backward & Exhaustive

Forward Selection

14 features

RMSE: 0.2805

Adds one variable at a time;
kept Governance & Social Score
but dropped Environmental

Backward Selection

12 features

RMSE: 0.2831

Starts with all, removes one at a time; kept only Governance
of the ESG pillars

Exhaustive Search

4 features

RMSE: 0.2807

Evaluates all subsets (capped at 4 for feasibility); found the optimal lean set

Why the Exhaustive Result Changes the Story

Exhaustive Search Selected:

Beta

Market Cap

EBITDA

Real Estate (sector dummy)

The Strategic Decision

We combined the exhaustive results with all three ESG pillar scores (Environmental, Social, Governance) to create a 7-feature set. This lets us directly compare ESG against proven financial metrics: the core question of our project.

Final 7-Feature Set for PCA & Machine Learning

Beta

Financial

Environmental Score

ESG

Market Cap

Financial

Social Score

ESG

EBITDA

Financial

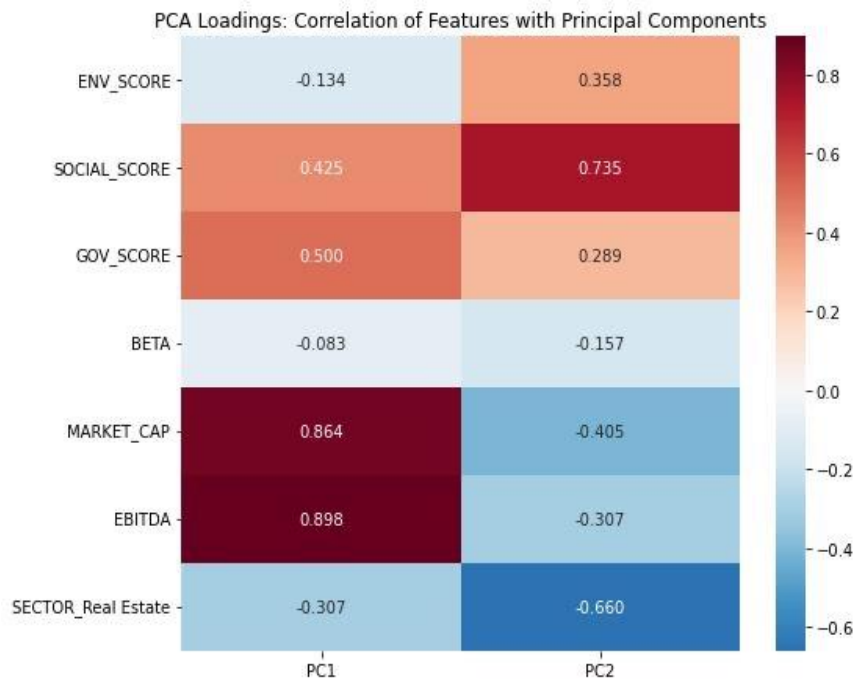
Governance Score

ESG

Real Estate (dummy)

Sector

PCA Loadings: What the Components Mean



PC1 — Company Size Axis

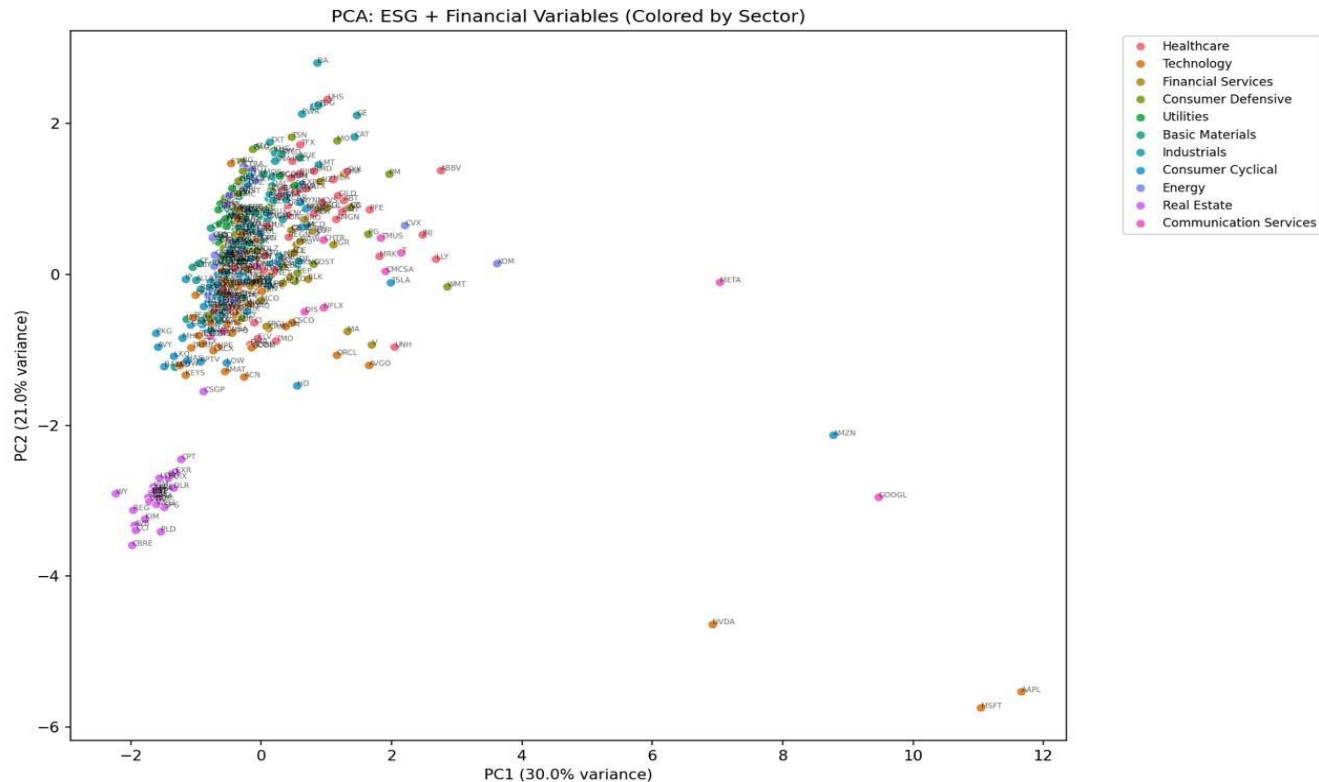
Dominated by Market Cap (0.86) and EBITDA (0.90). Governance and Social scores also load moderately — larger companies tend to score higher on governance.

PC2 — ESG Quality Axis

Driven by Social Score (0.74) and Environmental Score (0.36). Real Estate loads strongly negative (-0.66), confirming its uniquely different ESG profile.

PC1 + PC2 capture 51% of total variance across the 7 features.

PCA: Visualizing the Data in Two Dimensions



Key Observations

- PC1 (30%) captures company size — Market Cap & EBITDA dominate
- PC2 (21%) captures ESG quality — Social Score drives this axis
- Real Estate stocks cluster separately in the bottom-left — a distinct profile
- Mega-caps (AAPL, MSFT, NVDA) are extreme outliers on PC1

K-Means Clustering: 3 Distinct Groups



Cluster 0 — Mega-Caps

6 stocks

Highest EBITDA & Market Cap

Avg return: +67.5%

Cluster 1 — Broad Market

352 stocks

Average ESG & financials

Avg return: +12.6%

Cluster 2 — Real Estate

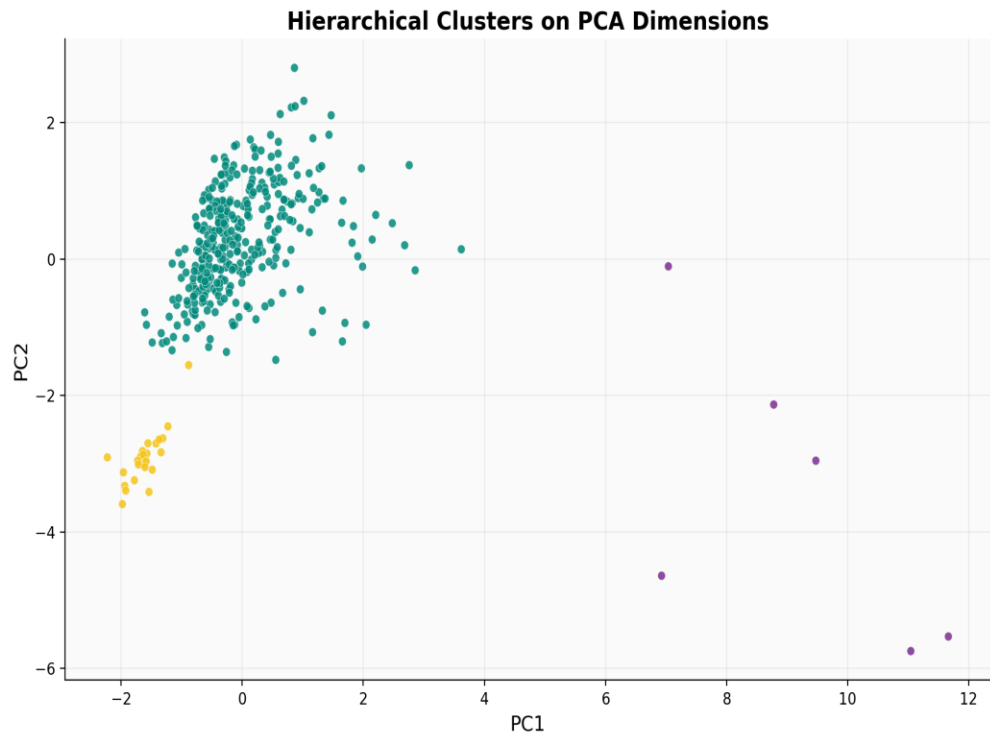
27 stocks

All Real Estate sector stocks

Avg return: -7.6%

Real Estate forms its own cluster with the worst performance — consistent with PCA findings.

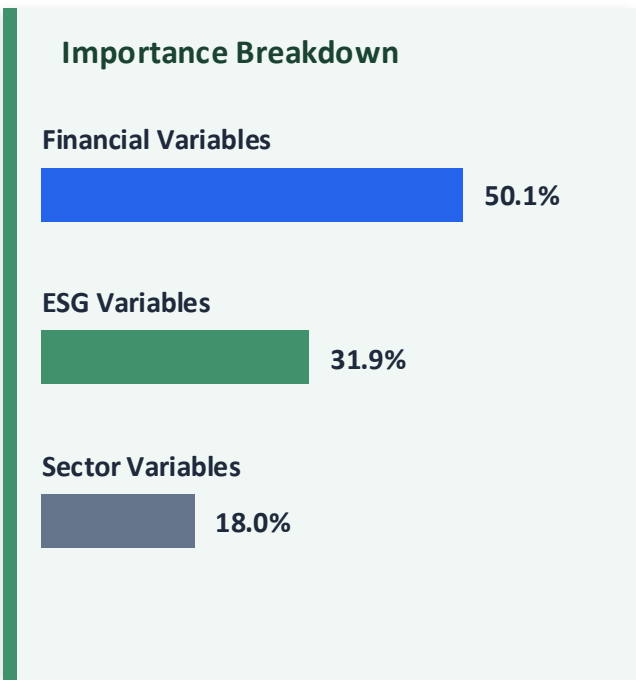
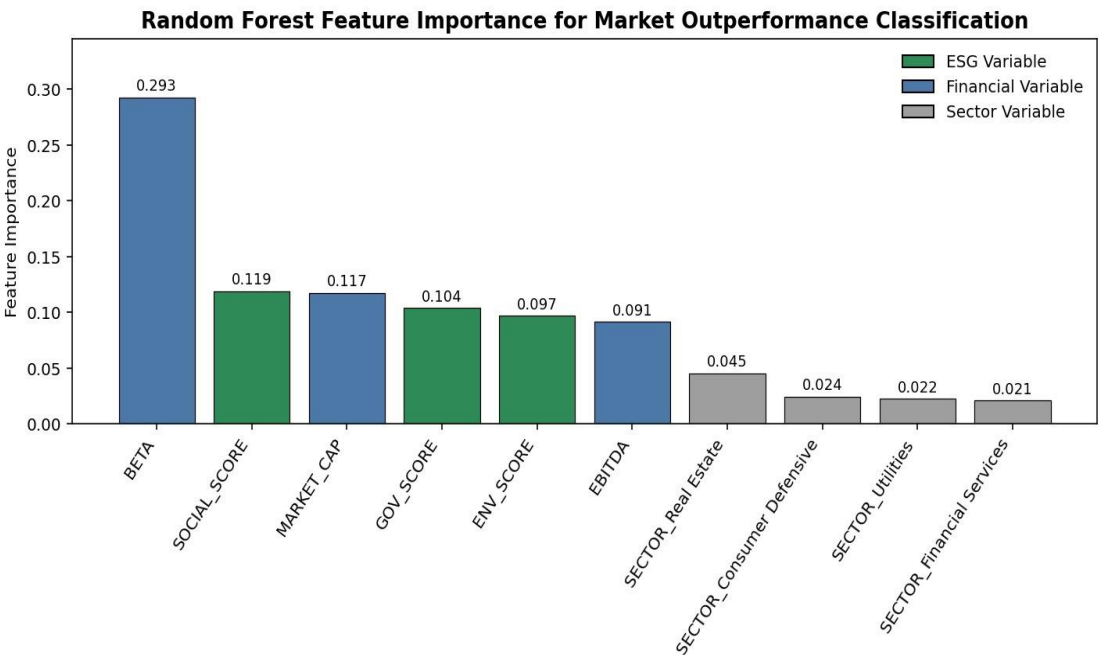
Hierarchical Clustering: Visual Confirmation



What This Confirms

- Same 3-group structure as K-Means, found independently
- Yellow cluster = Real Estate companies, isolated from every other sector
- Teal = mega-cap outliers (AAPL, MSFT, NVDA, GOOGL, AMZN, META)
- **Two algorithms, same conclusion: Real Estate is fundamentally distinct**

Random Forest: What Drives Outperformance?

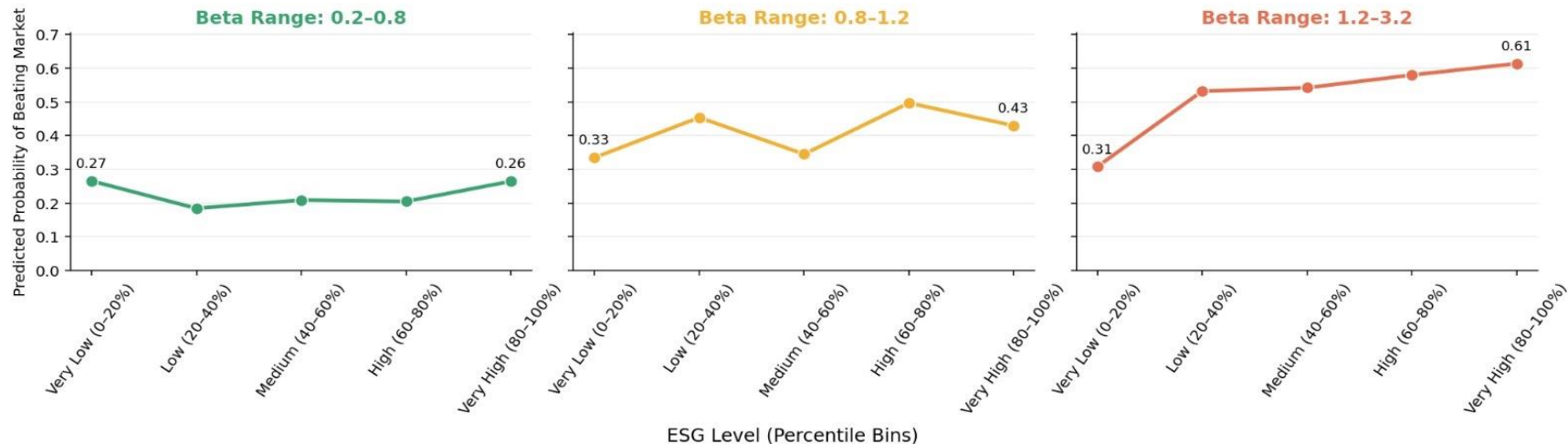


ROC-AUC: 0.81 | Accuracy: 70.1% | CV ROC-AUC: 0.76

Beta alone accounts for 29.3% of prediction power — financial risk dominates short-term returns.

Key Insight: ESG Amplifies High-Risk Returns

Model-Predicted Market Outperformance by ESG Level, Stratified by Beta



High-Beta + High ESG

61.4% predicted probability of beating the market: nearly double the 30.8% for low-ESG stocks in the same risk category.

Low-Beta + Any ESG

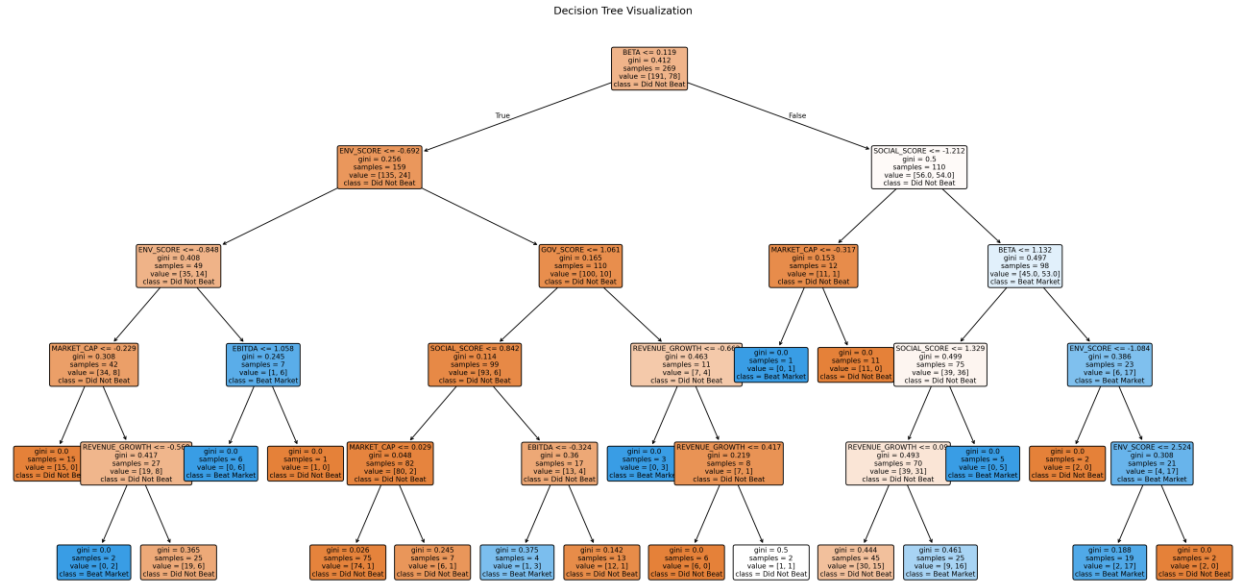
ESG makes almost no difference for low-risk stocks (~20-27% probability regardless of ESG level). ESG is not a standalone driver — it is a risk management amplifier.

Decision Tree: How did the model determine stock return?

Initially determines groups by Beta

Uses ESG to determine company type/priorities

Finally, segments companies based on financial metrics



Machine Learning: Model Comparison

Model	Accuracy	Precision (beat market)	Recall (beat market)	F1- score (beat market)
Linear Regression best accuracy	81.90%	0.50	0.33	0.40
Logistic Regression	79.31%	0.41	0.33	0.37
Random Forest Patrick	70.13%	0.43	0.45	0.44
Decision Tree	69.83%	0.31	0.52	0.39

S&P 500 benchmark return: 24% (Sep 2023 – Sep 2024) | Test set: 116 companies | 21 beat market, 95 did not

Target: Beat Market (1 Year Return > 24%) | Test: 116 companies | 21 beat, 95 did not

Conclusions: Why This Matters

1 Financial risk leads, but ESG follows closely

Beta is the #1 predictor at 29.3%, but ESG variables collectively contribute 31.9%; comparable to all financial variables combined (50.1%).

2 ESG is a risk management tool, not a standalone alpha generator

For high-Beta stocks, high ESG boosted predicted probability of beating market to 61% vs. 31% for low ESG — a near doubling. For low-Beta stocks, ESG barely moved the needle.

3 Real Estate is the canary in the ESG coal mine

PCA and clustering both isolated Real Estate with unique ESG profiles and the worst returns (-7.6%). This has implications for climate risk.

4 This is why firms like BlackRock factor ESG into decisions

ESG doesn't replace financial analysis, it enhances it. Our findings support ESG as a risk overlay, particularly for volatile, higher-risk investments.

Looking Forward: Real Estate, Climate Risk & ESG

The Real Estate - ESG Connection

Real Estate was uniquely isolated in its ESG profile and underperformed other sectors by a wide margin (-7.6% vs. +12.6% market average).

As natural disasters increase in frequency and severity, the real estate industry will be forced to develop more resilient, sustainable construction. This will intuitively drive higher Environmental scores across the sector over time.

Given our finding that ESG amplifies returns for higher-risk assets, improving environmental practices may not just be good ethics — it could close the performance gap and make Real Estate a more attractive investment.

The Bottom Line

ESG is no magic bullet for returns, but it is a meaningful risk management lever, especially for volatile sectors and high-Beta stocks. As climate pressures intensify, ESG integration will shift from optional to essential.

Thank You

Questions?